

# Commerce XI Maths App

## Exercise 3.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 17

### Question 1 - MCQ · PG-40 · EX-Q1(i)

Find the square root of complex number,  $-8 - 6i$

#### Options

- A.  $\pm(1 - 3i)$
- B.  $\pm(1 + 3i)$
- C.  $\pm(3 - i)$
- D.  $\pm(2 - 2i)$

**Correct answer:  $\pm(1 - 3i)$**

#### Explanation and steps

Let  $a + ib$  be a square root. Expand  $(a + ib)^2 = a^2 - b^2 + 2abi$ , equate real and imaginary parts, and include both signs. The roots are  $\pm(1 - 3i)$ .

### Question 2 - MCQ · PG-40 · EX-Q1(ii)

Find the square root of complex number,  $7 + 24i$

#### Options

- A.  $\pm(4 + 3i)$
- B.  $\pm(4 - 3i)$
- C.  $\pm(3 + 4i)$
- D.  $\pm(7 + 12i)$

**Correct answer:  $\pm(4 + 3i)$**

#### Explanation and steps

Let  $a + ib$  be a square root. Expand  $(a + ib)^2 = a^2 - b^2 + 2abi$ , equate real and imaginary parts, and include both signs. The roots are  $\pm(4 + 3i)$ .

### Question 3 - MCQ · PG-40 · EX-Q1(iii)

Find the square root of complex number,  $1 + 4\sqrt{3}i$

#### Options

- A.  $\pm(2 + \sqrt{3})i$
- B.  $\pm(2 - \sqrt{3})i$
- C.  $\pm(\sqrt{3} + 2i)$
- D.  $\pm(1 + 2\sqrt{3})i$

**Correct answer:  $\pm(2 + \sqrt{3})i$**

#### Explanation and steps

Let  $a + ib$  be a square root. Expand  $(a + ib)^2 = a^2 - b^2 + 2abi$ , equate real and imaginary parts, and include both signs. The roots are  $\pm(2 + \sqrt{3})i$ .

#### Question 4 - MCQ · PG-40 · EX-Q1(iv)

Find the square root of complex number,  $3 + 2\sqrt{10}i$

##### Options

A.  $\pm(\sqrt{5} + \sqrt{2}i)$

B.  $\pm(\sqrt{5} - \sqrt{2}i)$

C.  $\pm(\sqrt{10} + i)$

D.  $\pm(\sqrt{2} + \sqrt{5}i)$

**Correct answer:**  $\pm(\sqrt{5} + \sqrt{2}i)$

##### Explanation and steps

Let  $a + ib$  be a square root. Expand  $(a + ib)^2 = a^2 - b^2 + 2abi$ , equate real and imaginary parts, and include both signs. The roots are  $\pm(\sqrt{5} + \sqrt{2}i)$ .

#### Question 5 - MCQ · PG-40 · EX-Q1(v)

Find the square root of complex number,  $2(1 - \sqrt{3}i)$

##### Options

A.  $\pm(\sqrt{3} - i)$

B.  $\pm(\sqrt{3} + i)$

C.  $\pm(1 - \sqrt{3}i)$

D.  $\pm(2 - \sqrt{3}i)$

**Correct answer:**  $\pm(\sqrt{3} - i)$

##### Explanation and steps

Let  $a + ib$  be a square root. Expand  $(a + ib)^2 = a^2 - b^2 + 2abi$ , equate real and imaginary parts, and include both signs. The roots are  $\pm(\sqrt{3} - i)$ .

#### Question 6 - MCQ · PG-40 · EX-Q2(i)

Solve quadratic equation,  $8x^2 + 2x + 1 = 0$

##### Options

A.  $x = (-1 \pm \sqrt{7}i)/(8)$

B.  $x = (1 \pm \sqrt{7}i)/(8)$

C.  $x = (-1 \pm 7i)/(8)$

D.  $x = (-2 \pm \sqrt{7}i)/(8)$

**Correct answer:**  $x = (-1 \pm \sqrt{7}i)/(8)$

##### Explanation and steps

Apply the quadratic formula  $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = (-1 \pm \sqrt{7}i)/(8)$ .

### Question 7 - MCQ · PG-40 · EX-Q2(ii)

Solve quadratic equation,  $2x^2 - \sqrt{3}x + 1 = 0$

#### Options

- A.  $x = (\sqrt{3} \pm \sqrt{5} i)/(4)$
- B.  $x = (-\sqrt{3} \pm \sqrt{5} i)/(4)$
- C.  $x = (\sqrt{3} \pm \sqrt{5})/(4)$
- D.  $x = (\sqrt{3} \pm i)/(2)$

**Correct answer:**  $x = (\sqrt{3} \pm \sqrt{5} i)/(4)$

#### Explanation and steps

Apply the quadratic formula  $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = (\sqrt{3} \pm \sqrt{5} i)/(4)$ .

### Question 8 - MCQ · PG-40 · EX-Q2(iii)

Solve quadratic equation,  $3x^2 - 7x + 5 = 0$

#### Options

- A.  $x = (7 \pm \sqrt{11} i)/(6)$
- B.  $x = (-7 \pm \sqrt{11} i)/(6)$
- C.  $x = (7 \pm 11i)/(6)$
- D.  $x = (7 \pm \sqrt{11})/(6)$

**Correct answer:**  $x = (7 \pm \sqrt{11} i)/(6)$

#### Explanation and steps

Apply the quadratic formula  $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = (7 \pm \sqrt{11} i)/(6)$ .

### Question 9 - MCQ · PG-40 · EX-Q2(iv)

Solve quadratic equation,  $x^2 - 4x + 13 = 0$

#### Options

- A.  $x = 2 \pm 3i$
- B.  $x = -2 \pm 3i$
- C.  $x = 4 \pm 13i$
- D.  $x = 2 \pm i$

**Correct answer:**  $x = 2 \pm 3i$

#### Explanation and steps

Apply the quadratic formula  $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = 2 \pm 3i$ .

### Question 10 - MCQ · PG-40 · EX-Q3(i)

Solve quadratic equation,  $x^2 + 3ix + 10 = 0$

#### Options

- A.  $x = 2i$  or  $-5i$
- B.  $x = -2i$  or  $5i$
- C.  $x = 2$  or  $-5$
- D.  $x = 5i$  or  $-2i$

**Correct answer:**  $x = 2i$  or  $-5i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = 2i$  or  $-5i$ .

### Question 11 - MCQ · PG-40 · EX-Q3(ii)

Solve quadratic equation,  $2x^2 + 3ix + 2 = 0$

#### Options

- A.  $x = (i)/(2)$  or  $-2i$
- B.  $x = (-i)/(2)$  or  $2i$
- C.  $x = i$  or  $-i$
- D.  $x = (1)/(2)$  or  $-2$

**Correct answer:**  $x = (i)/(2)$  or  $-2i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = (i)/(2)$  or  $-2i$ .

### Question 12 - MCQ · PG-40 · EX-Q3(iii)

Solve quadratic equation,  $x^2 + 4ix - 4 = 0$

#### Options

- A.  $x = -2i, -2i$
- B.  $x = 2i, 2i$
- C.  $x = -4i$  or  $i$
- D.  $x = \pm 2i$

**Correct answer:**  $x = -2i, -2i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = -2i, -2i$ .

### Question 13 - MCQ · PG-40 · EX-Q3(iv)

Solve quadratic equation,  $ix^2 - 4x - 4i = 0$

#### Options

A.  $x = -2i, -2i$

B.  $x = 2i, 2i$

C.  $x = \pm 2i$

D.  $x = -4i$  or  $0$

**Correct answer:**  $x = -2i, -2i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = -2i, -2i$ .

### Question 14 - MCQ · PG-40 · EX-Q4(i)

Solve quadratic equation,  $x^2 - (2 + i)x - (1 - 7i) = 0$

#### Options

A.  $x = 3 - i$  or  $-1 + 2i$

B.  $x = 3 + i$  or  $-1 - 2i$

C.  $x = 2 + i$  or  $-1 + 7i$

D.  $x = 1 - i$  or  $-3 + 2i$

**Correct answer:**  $x = 3 - i$  or  $-1 + 2i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = 3 - i$  or  $-1 + 2i$ .

### Question 15 - MCQ · PG-40 · EX-Q4(ii)

Solve quadratic equation,  $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

#### Options

A.  $x = 3\sqrt{2}$  or  $2i$

B.  $x = -3\sqrt{2}$  or  $-2i$

C.  $x = \sqrt{2}$  or  $6i$

D.  $x = 3\sqrt{2}i$  or  $2$

**Correct answer:**  $x = 3\sqrt{2}$  or  $2i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = 3\sqrt{2}$  or  $2i$ .

### Question 16 - MCQ · PG-40 · EX-Q4(iii)

Solve quadratic equation,  $x^2 - (5 - i)x + (18 + i) = 0$

#### Options

A.  $x = 3 - 4i$  or  $2 + 3i$

B.  $x = 3 + 4i$  or  $2 - 3i$

C.  $x = 5$  or  $-i$

D.  $x = 3 - 2i$  or  $2 + i$

**Correct answer:**  $x = 3 - 4i$  or  $2 + 3i$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = 3 - 4i$  or  $2 + 3i$ .

### Question 17 - MCQ · PG-40 · EX-Q4(iv)

Solve quadratic equation,  $(2 + i)x^2 - (5 - i)x + 2(1 - i) = 0$

#### Options

A.  $x = 1 - i$  or  $(4 - 2i)/(5)$

B.  $x = 1 + i$  or  $(4 + 2i)/(5)$

C.  $x = 1$  or  $(1 - 2i)/(5)$

D.  $x = -1 - i$  or  $(2 - 4i)/(5)$

**Correct answer:**  $x = 1 - i$  or  $(4 - 2i)/(5)$

#### Explanation and steps

Apply the quadratic formula  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ , simplify negative radicals using  $i = \sqrt{-1}$ , and reduce. The solution is  $x = 1 - i$  or  $(4 - 2i)/(5)$ .